

Automated Parking Allocation System

5-Step Execution Flow & Interactive CLI Output Simulation

This document details the step-by-step program execution and output lifecycle of the parking script, showcasing how initialization, user inputs, allocation logic, payment handling, and termination occur.

Step 1: System Initialization & Menu Prompt

The system sets up a 2-floor matrix with 3 empty slots per floor and sets the standard rate to ₹20/hour. It then enters an infinite loop waiting for user choices.

CONSOLE MENU OUTPUT

```
==== PARKING SYSTEM ====
1. Book Slot
2. Exit & Pay
3. Show Parking
4. Exit
Enter choice:
```

Step 2: Viewing Initial Empty Layout (Choice 3)

The user inputs option 3 to view current allocations. The system loops through the empty data objects and prints the clean status layout.

CONSOLE INTERACTION & STATUS OUTPUT

```
Enter choice: 3

--- Parking Status ---

Floor 1:
  Slot 1: EMPTY
  Slot 2: EMPTY
  Slot 3: EMPTY

Floor 2:
  Slot 1: EMPTY
  Slot 2: EMPTY
  Slot 3: EMPTY
-----
```

Step 3: Allocating and Booking a Vehicle (Choice 1)

The user selects option 1. The program requests vehicle identification, customer name, and contact details, automatically matching them to the first free index (Floor 1, Slot 1) using `datetime.now()`.

CONSOLE INTERACTION & BOOKING RECEIPT

```
Enter choice: 1
Enter vehicle number: DL-3C-1234
Enter customer name: Alex
Enter phone number: 9876543210

✅ Slot booked!
Name: Alex
Phone: 9876543210
Vehicle: DL-3C-1234
Floor: 1, Slot: 1
Entry Time: 2026-05-29 16:53:24.123456
```

Step 4: Billing Calculation & Ticket Clearance (Choice 2)

The user processes vehicle departure via option 2. The manual ceiling logic evaluates the difference between entry and exit times. Any fractional duration (even a few minutes) automatically rounds up to the next full hour.

CONSOLE INTERACTION & BILL SUMMARY

```
Enter choice: 2
Enter vehicle number: DL-3C-1234
Payment Mode (UPI/Cash/Card): UPI

----- BILL -----
Name: Alex
Phone: 9876543210
Vehicle: DL-3C-1234
Floor: 1 Slot: 1
Entry: 2026-05-29 16:53:24.123456
Exit: 2026-05-29 16:55:12.789123
Hours: 1
Amount: ₹ 20
Payment Mode: UPI
-----
```

Step 5: Graceful Termination (Choice 4)

The system returns to the main menu dashboard. Selecting option 4 fires a break statement, safely freeing application memory and ending execution.

CONSOLE EXIT OUTPUT

```
Enter choice: 4
```

```
Exiting... 🖐️
```